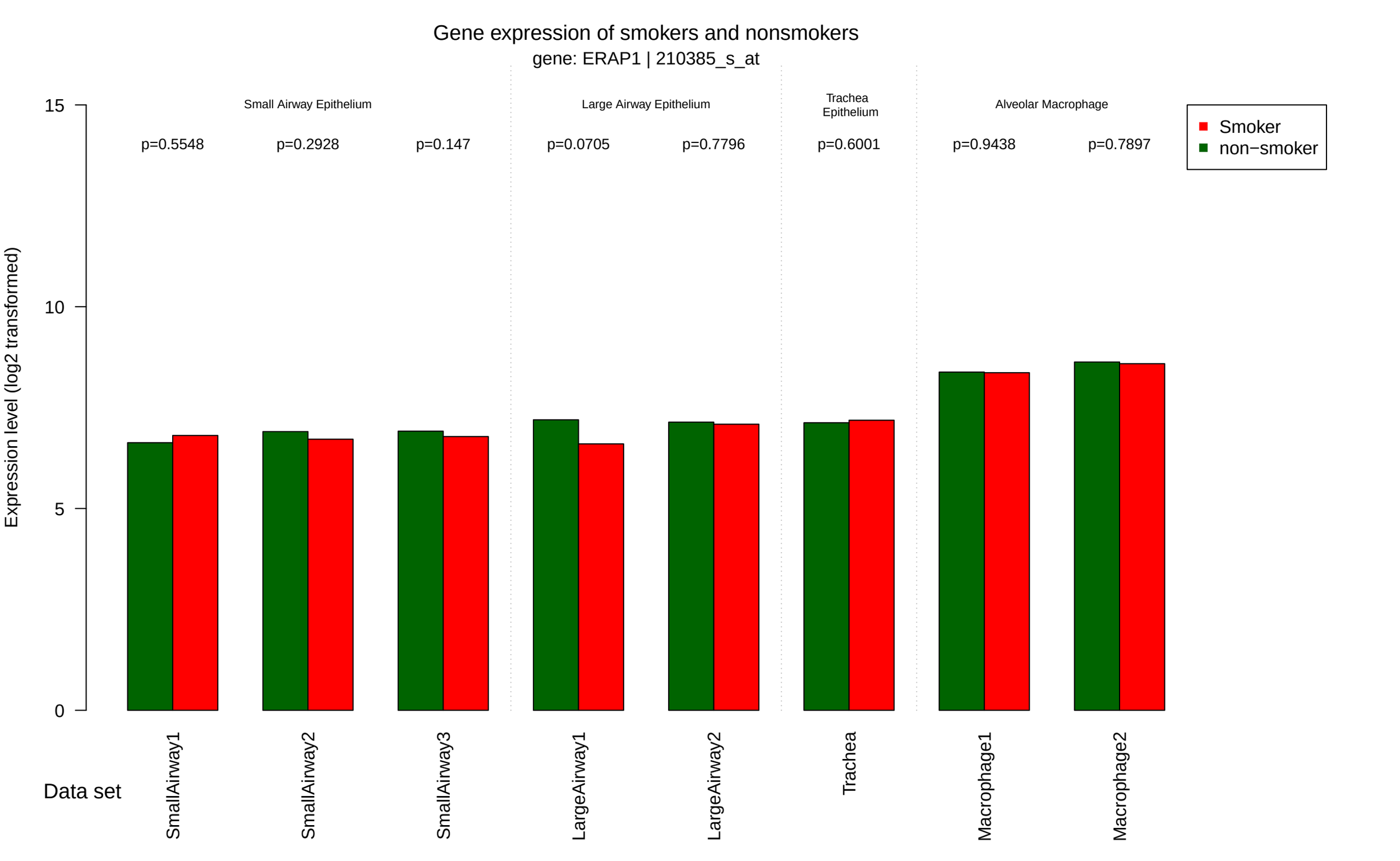
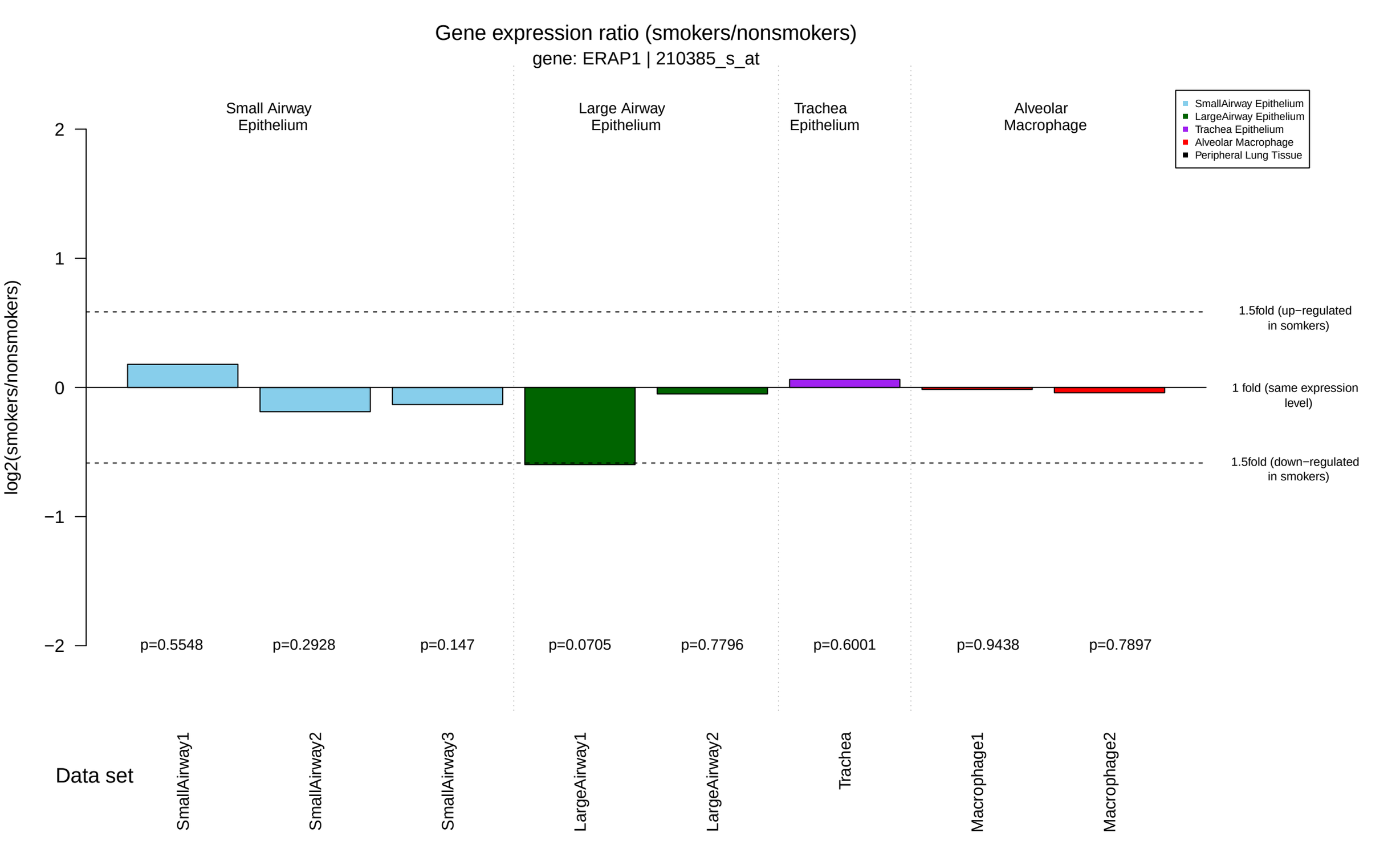
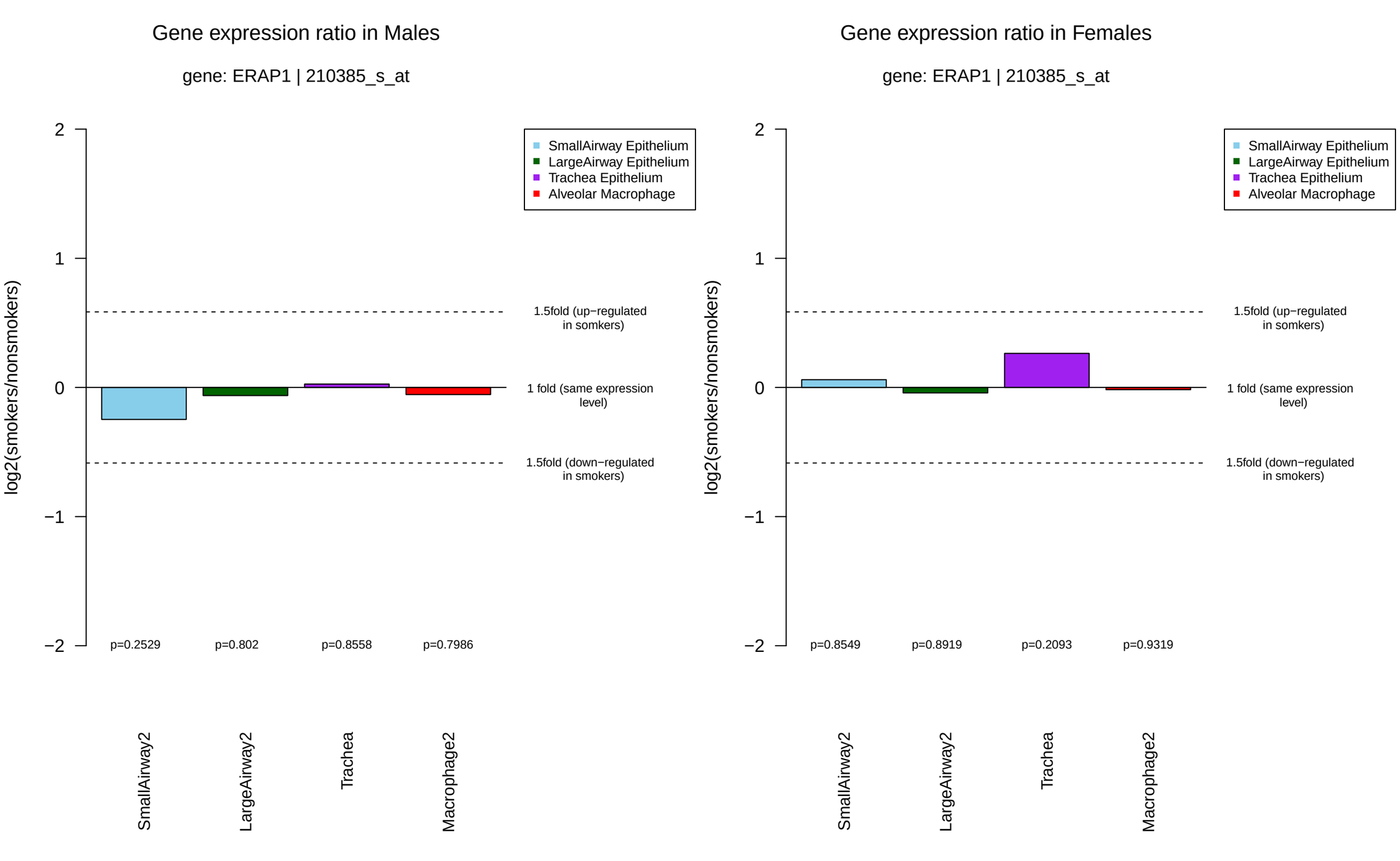
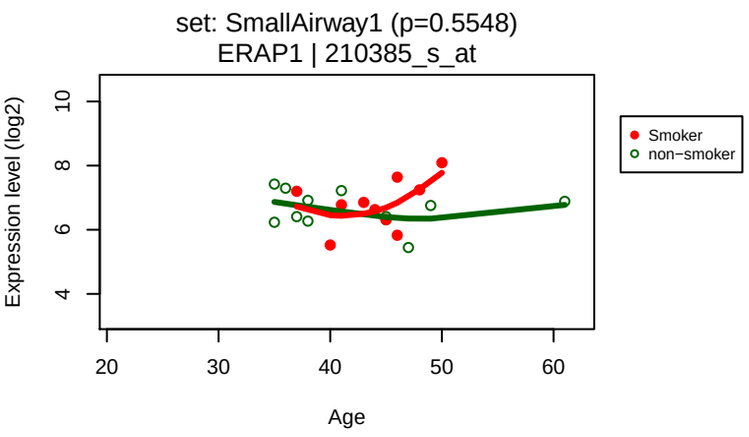
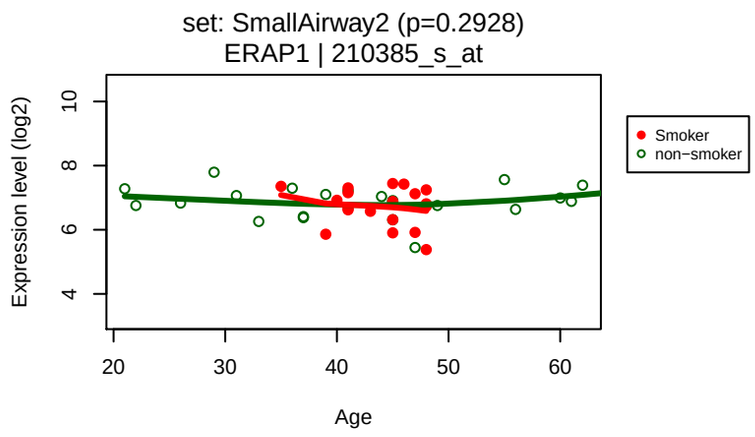
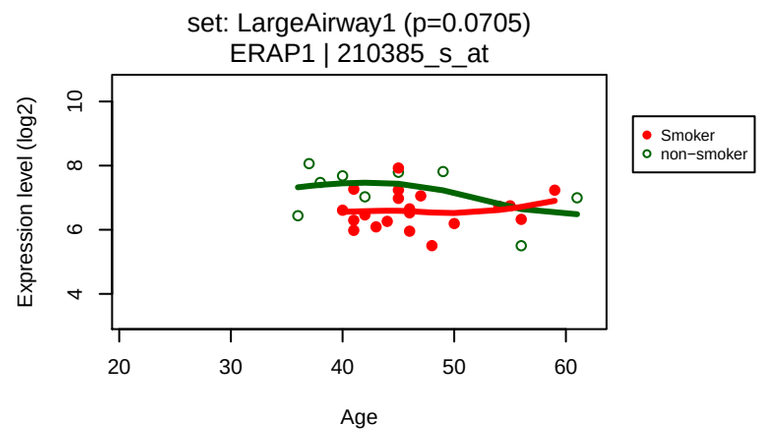

Gene expression of smokers and nonsmokers
gene: ERAP1 | 210385_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5548
p=0.2928
p=0.147
p=0.0705
p=0.7796
p=0.6001
p=0.9438
p=0.7897
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

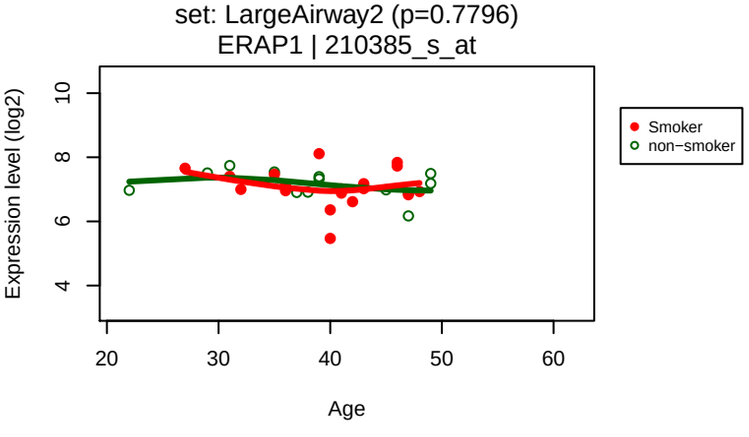

Gene expression ratio (smokers/nonsmokers)
gene: ERAP1 | 210385_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5548
p=0.2928
p=0.147
p=0.0705
p=0.7796
p=0.6001
p=0.9438
p=0.7897
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

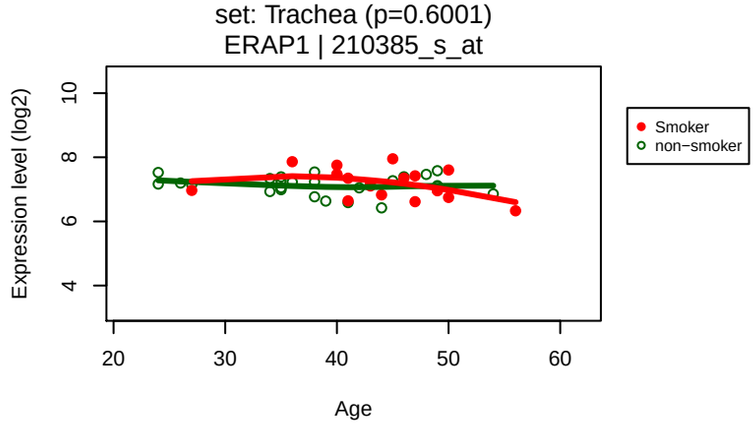

Gene expression ratio in Males
gene: ERAP1 | 210385_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2529
p=0.802
p=0.8558
p=0.7986
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ERAP1 | 210385_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8549
p=0.8919
p=0.2093
p=0.9319
SmallAirway2
LargeAirway2
Trachea
Macrophage2

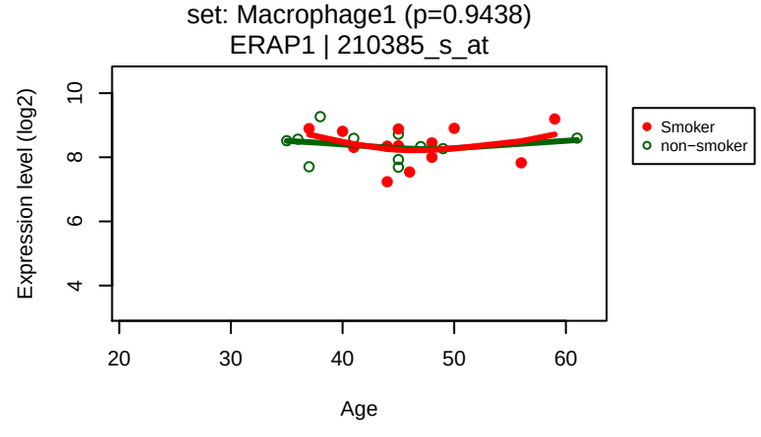

set: SmallAirway1 (p=0.5548)
ERAP1 | 210385_s_at
Expression level (log2)
Age
Smoker
non−smoker

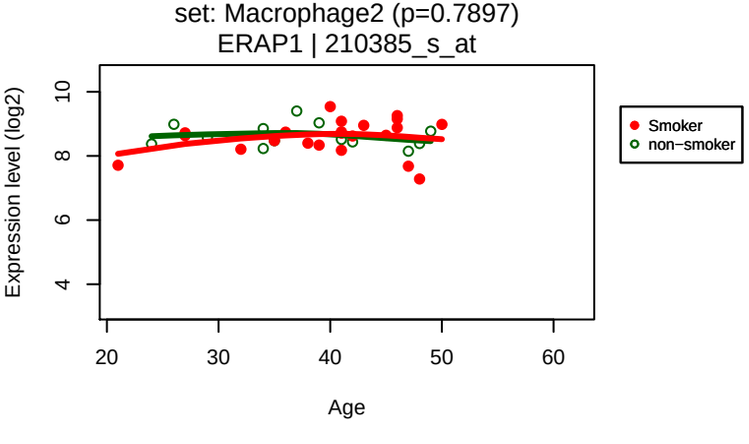

set: SmallAirway2 (p=0.2928)
ERAP1 | 210385_s_at
Expression level (log2)
Age
Smoker
non−smoker

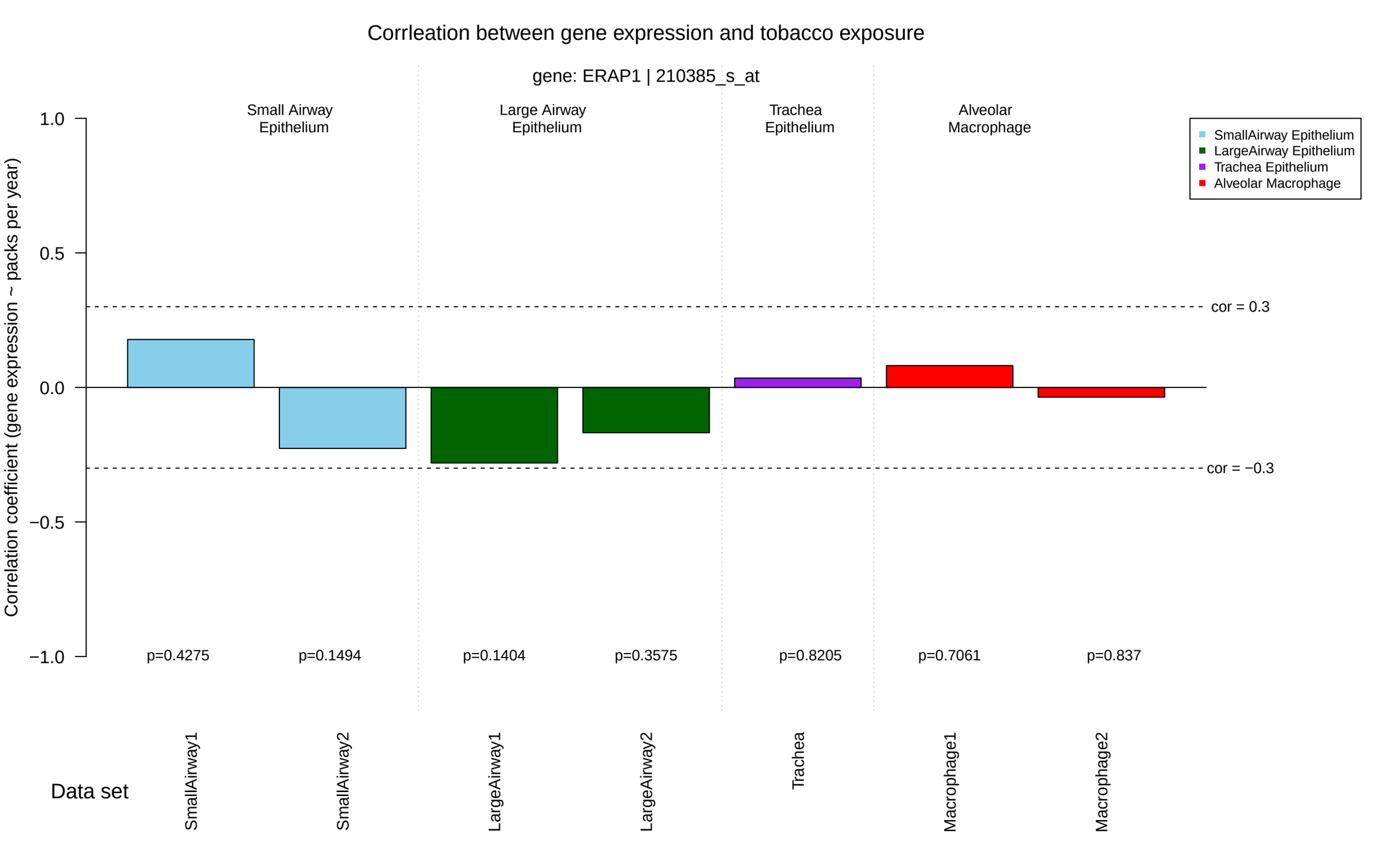

set: LargeAirway1 (p=0.0705)
ERAP1 | 210385_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7796)
ERAP1 | 210385_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6001)
ERAP1 | 210385_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9438)
ERAP1 | 210385_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7897)
ERAP1 | 210385_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ERAP1 | 210385_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4275
p=0.1494
p=0.1404
p=0.3575
p=0.8205
p=0.7061
p=0.837
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2